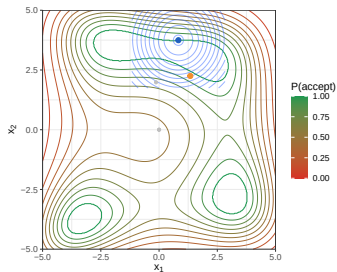
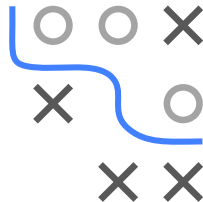


Optimization in Machine Learning

Simulated Annealing

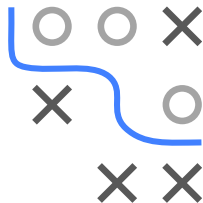


Learning goals

- Motivation
- Metropolis algorithm
- Simulated Annealing

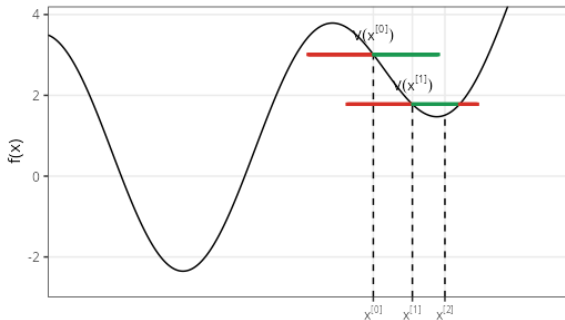
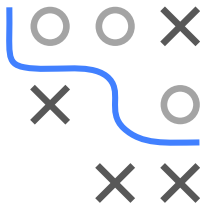
INTRODUCTION

- Heuristics for optimization of complex objectives (multivariate, non-linear, non-convex)
- Procedure for finding good solutions to complex problems
- Does not guarantee optimal/best result (global optimum), but usually good solutions
- Goal for complex optimization problems: avoid “getting stuck” in local optima
- Often used for difficult discrete problems as well
- Local search strategy with random option to accept worse values



SIMPLE STOCHASTIC LOCAL SEARCH

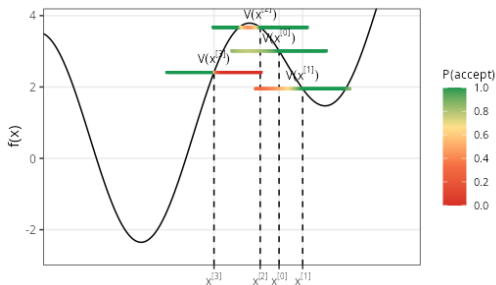
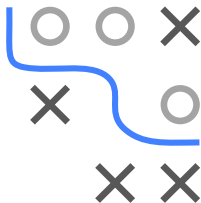
- Given is a multivariate objective function $f(\mathbf{x})$
- Define a local neighborhood area $V(\mathbf{x})$ for a given $\mathbf{x}$
- Sample proposal $\mathbf{x}^{[t+1]}$ uniformly at random from $V(\mathbf{x}^{[t]})$
- Calculate $f(\mathbf{x}^{[t+1]})$
- If $\Delta f = f(\mathbf{x}^{[t+1]}) - f(\mathbf{x}^{[t]}) < 0$, $\mathbf{x}^{[t+1]}$ is accepted as new solution, otherwise a new proposal from neighborhood is sampled



stoch. local search: acceptance range in green and rejection in red

METROPOLIS ALGORITHM

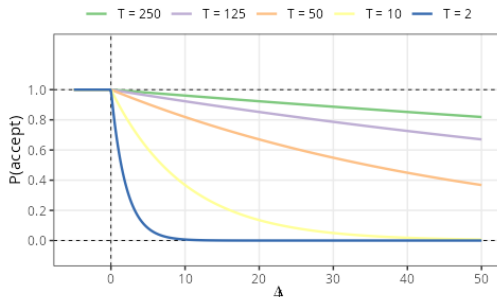
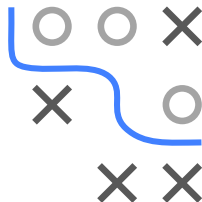
- Simple stochastic local search strongly depends on $\mathbf{x}^{[0]}$ and the neighborhood
⇒ Danger of ending up in local minima
- **Idea:** allow worse candidates with some probability
- **Metropolis:** accept candidates from previous rejection range ($\Delta f > 0$) with probability $\mathbb{P}(\text{accept} \mid \Delta f) = \exp(-\Delta f / T)$
- T denotes “temperature”



Simulated annealing: Colors correspond to $\mathbb{P}(\text{accept})$

METROPOLIS ALGORITHM

- Parameter T describes temperature/progress of the system
- High temperatures correspond to high probability of accepting worse x
- Local minima can be escaped, but no convergence can be achieved at *constant* temperature
- We come across an important principle of optimization:
exploration (high T) vs. exploitation (low T)



SIMULATED ANNEALING

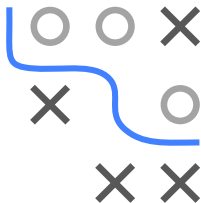
- Start with high temperature to explore whole space
- Slowly reduce temperature to converge
⇒ Sequence of descending temperatures $\mathcal{T}^{[t]}, t \in \mathbb{N}$
- Procedure is called simulated annealing
- Temperature is often kept constant several iterations in a row to explore the space, then multiplied by coefficient $0 < c < 1$:

$$\mathcal{T}^{[t+1]} = c \cdot \mathcal{T}^{[t]}$$

- Other strategies possible, for example:

$$\mathcal{T}^{[t]} = \mathcal{T}^{[0]} \left(1 - \frac{t}{t_{\max}} \right)$$

- Many different strategies for choosing neighborhood
Strongly depends on objective function



ANALOGY TO METALLURGY

- Simulated annealing draws analogy between a cooling process (e.g. a metal or liquid) and an optimization problem.
- If cooling of a liquid material (amount of atoms) is too fast, it solidifies in suboptimal configuration, slow cooling produces crystals with optimal structure (minimum energy stage)
- Consider atoms of the liquid as a system with many degrees of freedom, analogy to optimization problem of a multivariate function
- Minimum energy stage corresponds to optimum of objective function

